Axial Holonomy and Visibility-Weighted Coherence: Cosmic Birefringence in the Relativistic Scalar–Vector Plenum

Flyxion

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Abstract

Cosmic birefringence, the rotation of CMB polarization by an angle $\beta \sim 0.3^\circ$ detected at high significance in recent data [5–7], is typically attributed to a Chern–Simons coupling between photons and axion-like particles (ALPs) [2, 3]. Here we reinterpret this phenomenon within the Relativistic Scalar–Vector Plenum (RSVP) as a *visibility-weighted axial holonomy* of a parity-odd sector of the plenum connection. The polarization rotation angle is the integral of an effective axial connection κ_A along null geodesics; the washout suppression of polarization amplitude arises from phase-variance of κ_A across the finite recombination visibility kernel $W(\eta)$ [9, 10]. We derive a transport formalism in which mean rotation and depolarization are governed respectively by the first and second spectral moments of the axial connection. The two-field ALP mechanism of Shao–Obata–Zhang [1] emerges as a special case of a multi-modal axial sector whose independent phase components factorize under visibility averaging, yielding the familiar $\sqrt{N}$ relaxation of washout constraints through variance additivity. A continuous synchronization model for the parity-odd phase field leads to a *coherence theorem*: the CMB polarization amplitude imposes a lower bound on the axial correlation length at $z \approx 1100$. This reframing replaces particle-specific exclusion statements with a geometric coherence inequality, reveals structural equivalence between cosmological birefringence and phase-stable photonic computation [42, 43], and predicts discriminants absent in minimal axion models, including correlations between birefringence anisotropies and entropy gradients, vorticity statistics, and multipole-dependent depolarization. Cosmic birefringence is thus recast as a probe of axial synchronization in the universe's parity-odd sector rather than solely as evidence for a propagating pseudoscalar field.

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1 Introduction

Recent high-significance measurements of cosmic microwave background (CMB) polarization have revived interest in the possibility of a nonzero global birefringence angle. The Planck, WMAP, and ACT data-sets together indicate a rotation $\beta \approx 0.35^\circ$ at roughly 3σ significance [5–8], with anisotropic components beginning to be resolved at degree angular scales. Conventionally, such rotation is modeled through a Chern–Simons coupling

$$\mathcal{L}_{\rm CS} = -\frac{g_{\phi\gamma}}{4}\,\phi\, F_{\mu\nu}\widetilde{F}^{\mu\nu}, \tag{1}$$

which leads to a polarization angle proportional to the ALP field displacement between emission and observation [2–4, 12]. Within that framework, the observational bound on polarization amplitude constrains the oscillation rate and coupling of the pseudoscalar through a washout condition arising from the finite thickness of recombination [9–11].

While this particle-based interpretation is algebraically sufficient, it is not logically necessary. The observable quantity is a rotation of the photon's transverse polarization frame, accumulated along null geodesics and weighted by the visibility function of recombination [13, 14]. At a structural level this is a *holonomy problem*: polarization is parallel-transported by an effective optical connection, and birefringence measures the parity-odd component of that connection integrated along the path. The washout bound constrains not a field species per se but the *phase coherence* of whatever geometric sector sources that parity-odd projection across the emission kernel.

In this paper we develop that geometric interpretation within the Relativistic Scalar–Vector Plenum (RSVP) framework. RSVP posits that cosmological dynamics are governed not solely by a metric tensor but by a coupled scalar–vector–entropy system (Φ, v^μ, S) defined on spacetime [17–19]. Gravitational and optical phenomena emerge from the combined geometry of these fields through effective connections constructed from their gradients and contractions. Parity-odd structures arise naturally from pseudoscalar combinations such as $\varepsilon_{\mu\nu\rho\sigma}k^\mu u^\nu \nabla^\rho S\, v^\sigma$, where k^μ is the photon wavevector and u^μ a comoving four-velocity [21,22].

Main results.

1. We derive an *axial transport law* (section 2) expressing CMB polarization rotation as the holonomy of a parity-odd sector of the RSVP plenum connection.
2. We establish a *visibility-weighted holonomy formalism* (section 3) that identifies the washout bound with a spectral coherence constraint on κ_A.
3. We show that the two-field ALP mechanism of [1] is a *special case* (section 4) of multi-modal axial coherence governed by variance additivity.

4. We prove a *coherence theorem* (section 6 and **??**) bounding the axial correlation length at recombination.
5. We derive an *entropy–coherence inequality* (section 8) unifying cosmological washout with the thermodynamic cost of phase-stable computation.
6. We identify *observational discriminants* (section 9) that distinguish geometric axial coherence from free pseudoscalar transport.

The paper is organized as follows. Section 2 develops the RSVP optical connection and axial holonomy. Section 3 establishes the visibility-weighted formalism and coherence constraints. Section 4 treats multi-modal factorization and the two-field mechanism. Section 5 discusses foliation structure and local-field dominance. Section 6 introduces the 5D Ising synchronization model. Section 7 derives the continuous axial phase model and states the coherence theorem. Section 8 establishes the entropy–coherence inequality and its implications for photonic computing. Section 9 presents observational discriminants. Section 10 synthesizes the results. Technical derivations occupy Appendices A–F.

2 Geometric Structure of the Relativistic Scalar–Vector Plenum

2.1 Plenum fields and the RSVP connection

The Relativistic Scalar–Vector Plenum (RSVP) is a geometric field theory in which spacetime carries three interlocking fields: a scalar density Φ, a vector flow v^μ, and an entropy density S. The metric $g_{\mu\nu}$ is interpreted as an emergent large-scale descriptor of averaged plenum structure rather than a primitive background variable [18–20]. Deviations from Levi–Civita transport arise from gradients and vorticity of (Φ, v^μ, S).

The full plenum connection is

$$\widetilde{\Gamma}^\mu{}_{\nu\rho} = \Gamma^\mu{}_{\nu\rho} + \Delta^\mu{}_{\nu\rho}(\Phi, v, S), \tag{2}$$

where $\Delta^\mu{}_{\nu\rho}$ represents non-Riemannian structure induced by plenum ordering [24, 25]. The correction decomposes irreducibly under spatial inversion into parity-even and parity-odd sectors. The parity-odd sector, which we denote $\Delta^{(\mathrm{odd})}$, acts differently on left- and right-handed polarization states; it is the geometric locus of cosmic birefringence in RSVP.

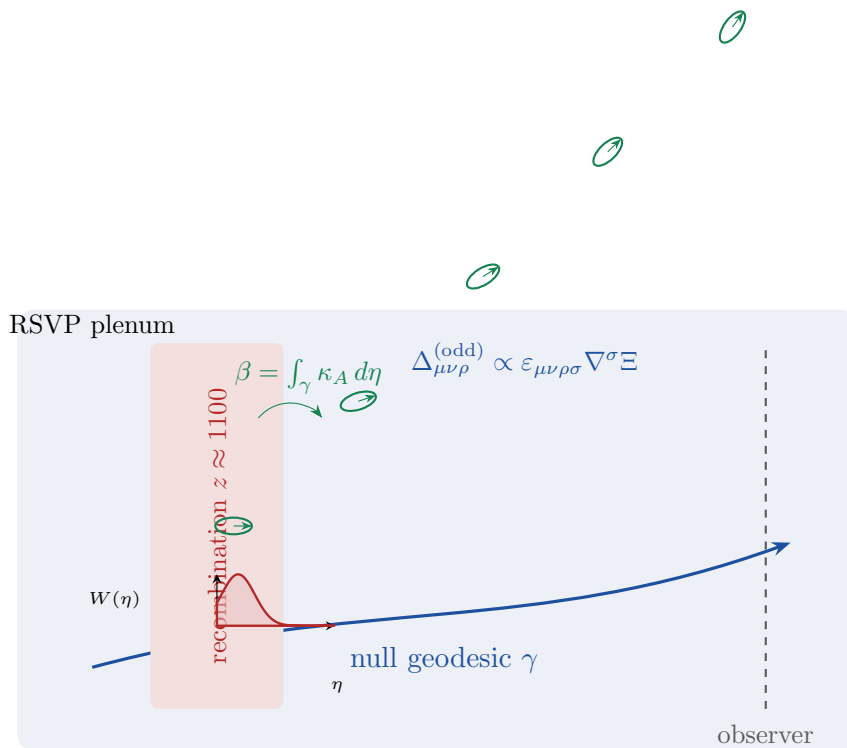

Figure 1: Schematic of axial holonomy along a null geodesic γ in the RSVP plenum. Photon polarization (green ellipses) rotates by an angle $\beta = \int_\gamma \kappa_A \, d\eta$ controlled by the parity-odd projection κ_A of the plenum connection $\Delta^{(\mathrm{odd})}$. The red band marks the recombination visibility kernel $W(\eta)$; phase oscillations faster than $\Delta\eta_{\mathrm{rec}}$ contribute to depolarization rather than coherent rotation.

2.2 Polarization transport equation

Let k^μ be the null tangent to γ and P^μ a transverse polarization vector with $P^\mu k_\mu = 0$. Standard geometric optics gives $k^\nu \nabla_\nu P^\mu = 0$ [26–28]; in RSVP the transport is instead

$$k^\nu \widetilde{\nabla}_\nu P^\mu = 0, \tag{3}$$

where $\widetilde{\nabla}$ uses the full plenum connection (2). Projecting onto the two-dimensional screen bundle $\{e_1^\mu, e_2^\mu\}$ orthogonal to k^μ and u^μ, the parity-odd sector induces an effective scalar angular velocity

$$\frac{d\theta}{d\lambda} = \kappa_A(\lambda), \qquad \kappa_A(\lambda) = \Pi^\mu{}_a \, \Pi^\nu{}_b \, \epsilon^{ab} \, k^\rho \Delta^{(\mathrm{odd})}_{\mu\nu\rho}, \tag{4}$$

so that the accumulated axial holonomy along γ is

$$\boxed{H_A[\gamma] = \int_{\eta_{\mathrm{LSS}}}^{\eta_0} \kappa_A(\eta, x(\eta)) \, d\eta.} \tag{5}$$

This is structurally identical to the Chern–Simons result $\beta = \frac{1}{2} g_{\phi\gamma}(\phi_{\eta_0} - \phi_{\eta_{\rm LSS}})$ under the identification

$$\kappa_A \longleftrightarrow \tfrac{1}{2} g_{\phi\gamma}\, \partial_\eta \phi, \tag{6}$$

but now κ_A is a projection of plenum geometry rather than a derivative of a particle field.

2.3 Internal origin of the parity-odd sector

A minimal covariant parity-odd connection compatible with RSVP fields is

$$\Delta^{(\rm odd)}_{\mu\nu\rho} = \alpha\, \varepsilon_{\mu\nu\rho\sigma}\, \nabla^\sigma \Xi(\Phi, v, S), \tag{7}$$

where α is a coupling scale and Ξ is any pseudoscalar functional of the plenum. The canonical choice coupling entropy gradients to vorticity is

$$\Xi = \nabla_\mu S\, \omega^\mu, \qquad \omega^\mu = \varepsilon^{\mu\nu\rho\sigma} u_\nu \nabla_\rho v_\sigma, \tag{8}$$

which is manifestly parity-odd and depends only on plenum kinematics, not on any new fundamental field [21–23]. The axion model is recovered when $\Xi \to \phi$ is promoted to an independent dynamical pseudoscalar with the action (1). RSVP treats Ξ as emergent from plenum ordering.

3 Visibility-Weighted Holonomy and Coherence Constraints

3.1 Observable polarization

CMB polarization is generated across a finite conformal time interval centered at recombination [13–15]. Let $W(\eta)$ be the visibility function, normalized to unity, with characteristic width $\Delta\eta_{\rm rec} \approx 10\,{\rm Mpc}/h$ at $z \approx 1100$ [59]. Defining complex polarization $P_\pm = Q \pm iU$, the observed polarization is

$$P^{\rm obs}_\pm(\hat{n}) = \int d\eta\, W(\eta)\, e^{\pm 2i\beta(\eta,\hat{n})}\, P^{(0)}_\pm(\eta, \hat{n}), \tag{9}$$

where $\beta(\eta, \hat{n}) = \int_\eta^{\eta_0} \kappa_A(\eta', x(\eta'))d\eta'$ is the holonomy from emission time η to the observer. Factoring out the slowly varying intrinsic polarization,

$$P^{\rm obs}_\pm = \mathcal{W}_\pm(\hat{n})\, P^{(0)}_\pm(\eta_\star, \hat{n}), \quad \mathcal{W}_\pm(\hat{n}) = \int d\eta\, W(\eta)\, e^{\pm 2i\beta(\eta,\hat{n})}. \tag{10}$$

The washout effect is $|\mathcal{W}_\pm| < 1$ due to phase decoherence across the kernel [9–11].

3.2 Spectral decomposition and Bessel suppression

Decompose κ_A into a coherent component and oscillatory modes:

$$\kappa_A(\eta) = \kappa_0(\eta) + \sum_k a_k \cos(\omega_k \eta + \varphi_k). \tag{11}$$

The holonomy becomes

$$\beta(\eta) = \bar{\beta} + \sum_k A_k \sin(\omega_k \eta + \varphi_k), \qquad A_k = \frac{a_k}{\omega_k}, \tag{12}$$

where $\bar{\beta} = \int_{\eta_\star}^{\eta_0} \kappa_0 d\eta'$. For modes with $\omega_k \Delta\eta_{\rm rec} \gg 1$ the integral over $W(\eta)$ reduces to a Bessel-J_0 suppression [47, 48]:

$$\int d\eta\, W(\eta)\, e^{iA_k \sin(\omega_k \eta)} \approx J_0(A_k). \tag{13}$$

With spectrally separated modes, the full suppression factorizes:

$$\mathcal{W}_\pm \approx e^{\pm 2i\bar{\beta}} \prod_k J_0(2A_k). \tag{14}$$

Expanding for small A_k:

$$|\mathcal{W}_\pm| \approx 1 - 2\sum_k A_k^2 + \mathcal{O}(A_k^4), \tag{15}$$

yielding the fundamental *coherence inequality*

$$\boxed{\sum_k A_k^2 \lesssim \epsilon_{\rm wash},} \tag{16}$$

where $\epsilon_{\rm wash}$ is the observationally determined upper bound on depolarization [5, 6, 10]. This bounds the spectral variance of κ_A at frequencies exceeding $\Delta\eta_{\rm rec}^{-1}$, not the mean coherent component $\bar{\beta}$.

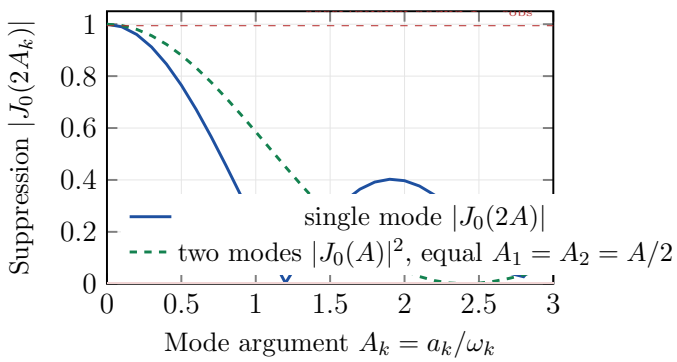

Figure 2: Washout suppression factor as a function of mode amplitude A_k. Blue: single-mode Bessel factor $|J_0(2A)|$. Green dashed: two-mode product $|J_0(A)|^2$ for equal-amplitude modes each contributing $A_k = A/2$. The two-mode configuration achieves the same mean rotation at reduced depolarization. The red dashed line marks the observational amplitude bound $|\mathcal{W}_\pm| \geq 0.9942$ from [10].

4 Multi-Modal Axial Connection and Factorized Dephasing

4.1 Two-field mechanism as special case

In the Shao–Obata–Zhang scenario [1], two ALP fields ϕ_1, ϕ_2 with common photon coupling contribute

$$\kappa_A(\eta) = -\tfrac{1}{2} g_{\phi\gamma} \sum_{i=1}^{2} \Phi_i m_i \sin(m_i \eta + \alpha_i), \tag{17}$$

with mode amplitudes $A_i = \frac{1}{2} g_{\phi\gamma} \Phi_i$ and frequencies $\omega_i = m_i$. When $|m_1 - m_2| \gg \Delta\eta_{\rm rec}^{-1}$, cross-terms average out and

$$\mathcal{W}_\pm \approx e^{\pm 2i\bar{\beta}}\, J_0(2A_1)\, J_0(2A_2). \tag{18}$$

This is the RSVP multi-modal result (14) for $k = 1, 2$: the two-field mechanism is a special instance of a two-mode axial sector.

4.2 Variance additivity and the $\sqrt{N}$ relief

For small amplitudes, the distinction between single- and two-mode washout is algebraic:

$$\text{single mode:} \quad J_0(2A) \approx 1 - 2A^2, \quad A = A_1 + A_2, \tag{19}$$

$$\text{two modes:} \quad J_0(2A_1) J_0(2A_2) \approx 1 - 2(A_1^2 + A_2^2). \tag{20}$$

The single-mode suppression at fixed total amplitude scales as $(A_1 + A_2)^2$; the two-mode product scales as $A_1^2 + A_2^2$. The difference $2A_1A_2$ is the cross-term that vanishes when the modes oscillate incoherently across the visibility kernel. For equal amplitudes $A_1 = A_2 \equiv \bar{A}$,

$$\frac{\text{two-mode washout}}{\text{single-mode washout}} = \frac{2\bar{A}^2}{4\bar{A}^2} = \tfrac{1}{2}, \tag{21}$$

giving a $\sqrt{2}$ improvement in coherent rotation per unit depolarization budget. For N equal-amplitude modes:

$$\frac{\langle\beta\rangle}{\sqrt{\mathrm{Var}_W[\Theta]}} \propto \sqrt{N}. \tag{22}$$

This is not a special property of axions but a *universal consequence of variance additivity under independent phase modes* [45, 46].

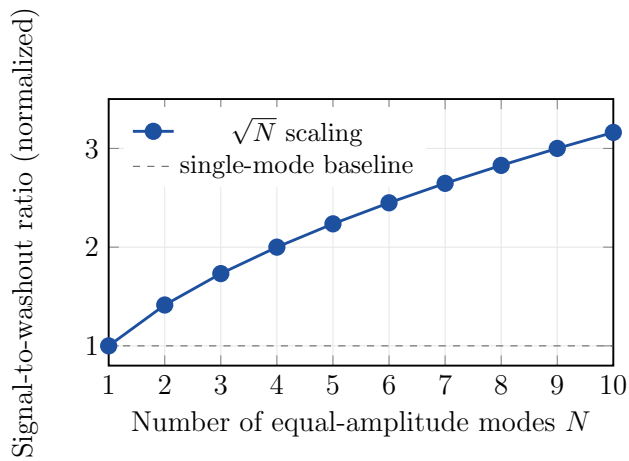

Figure 3: Signal-to-washout ratio $\langle\beta\rangle/\sqrt{\mathrm{Var}_W[\Theta]}$ normalized to unity for $N = 1$, showing universal $\sqrt{N}$ scaling from variance additivity under N independent axial modes. The two-field mechanism of [1] corresponds to $N = 2$.

5 Boundary Dominance and Foliation Structure

5.1 Origin of the boundary-term structure

The standard axion result $\beta = \frac{1}{2}g_{\phi\gamma}(\phi_{\eta_0} - \phi_{\eta_{\mathrm{LSS}}})$ is a boundary term. Within RSVP this structure is not an artifact of the axion equation of motion but a *structural consequence of visibility-weighted holonomy.*

Interchanging integration order in $\beta_{\rm obs} = \int W(\eta)\beta(\eta)d\eta$:

$$\beta_{\rm obs} = \int \kappa_A(\eta')\,\mathcal{F}(\eta')\,d\eta', \qquad \mathcal{F}(\eta') = \int_{\eta_{\rm LSS}}^{\eta'} W(\eta)\,d\eta. \tag{23}$$

The cumulative visibility $\mathcal{F}(\eta')$ rises sharply from 0 to 1 across $\Delta\eta_{\rm rec}$ and remains unity thereafter. For oscillatory modes with $\omega_k\Delta\eta_{\rm rec} \gg 1$, the emission-window contribution $\int_{\eta_{\rm LSS}}^{\eta_\star} \kappa_A\,\mathcal{F}\,d\eta'$ averages to zero, leaving

$$\beta_{\rm obs} \approx \int_{\eta_\star}^{\eta_0} \kappa_A(\eta')\,d\eta', \tag{24}$$

a *post-recombination* integral depending on the observer-side plenum state [9, 11].

5.2 Local-field dominance as a structural fact

When modes oscillate many times between $\eta_\star$ and η_0, the LSS contribution $\int^{\eta_\star} \kappa_A\,d\eta'$ is effectively a random phase in the ensemble:

$$\left\langle \int_{\eta_{\rm LSS}}^{\eta_\star} \kappa_A(\eta')\,\mathcal{F}(\eta')\,d\eta' \right\rangle = 0. \tag{25}$$

The observed isotropic β measures the axial state of the plenum at the *observer's foliation*—a single evaluation point—not an integrated history. This makes local-field dominance not mysterious but structural: it is what any ordering-based observable does when the source kernel is broad relative to the medium's oscillation period [10, 13].

6 A Five-Dimensional Ising Synchronization Model

The coherence constraint (16) motivates a constructive question: what class of mesoscopic dynamics can generate a parity-odd sector with tunable correlation length? We introduce a five-dimensional Ising-type synchronization model to answer this.

6.1 Lattice construction

Let $\Lambda \subset \mathbb{Z}^5$ be a hypercubic lattice with sites $x = (n_0, n_1, n_2, n_3, n_4)$, where the first four coordinates represent a coarse-grained comoving spacetime lattice and n_4 indexes an internal axial layer. At each site we place an Ising spin $\sigma_x \in \{-1, +1\}$ representing the sign of the local parity-odd plenum response [29, 30].

The anisotropic Hamiltonian is

$$\mathcal{H}[\sigma] = -\sum_{x\in\Lambda}\left(\sum_{\mu=0}^{3} J_\mu\,\sigma_x\sigma_{x+\hat{\mu}} + J_4\,\sigma_x\sigma_{x+\hat{4}}\right) - \sum_{x\in\Lambda} h_x\sigma_x, \tag{26}$$

where J_4 controls axial-layer synchronization and $h_x \propto \Xi(x)$ encodes slow RSVP biases from the plenum pseudoscalar (8).

The coarse-grained axial integrand along a null ray is

$$\kappa_A(\eta) = \kappa_0\, m(\eta), \qquad m(\eta) = \mathbb{E}[\sigma_{x(\eta)}], \tag{27}$$

with correlation function

$$C(\Delta\eta) = \mathbb{E}[m(\eta)m(\eta+\Delta\eta)] - \mathbb{E}[m]^2. \tag{28}$$

6.2 Washout as a correlation-length constraint

For exponential decay $C(\Delta\eta) \sim e^{-|\Delta\eta|/\xi_{\rm rec}}$, the visibility-weighted variance scales as

$$\mathrm{Var}_W[\Theta] \sim 4\kappa_0^2 \int d\eta\, d\eta'\, W(\eta)W(\eta')\, e^{-|\eta-\eta'|/\xi_{\rm rec}}. \tag{29}$$

The qualitative regime distinction is immediate (see fig. 4):

- $\xi_{\rm rec} \gg \Delta\eta_{\rm rec}$: coherent sector; exponential ≈ 1 over $\mathrm{supp}(W)$; small variance, large mean rotation.
- $\xi_{\rm rec} \ll \Delta\eta_{\rm rec}$: incoherent sector; variance extensive in sub-intervals; strong depolarization, suppressed mean.

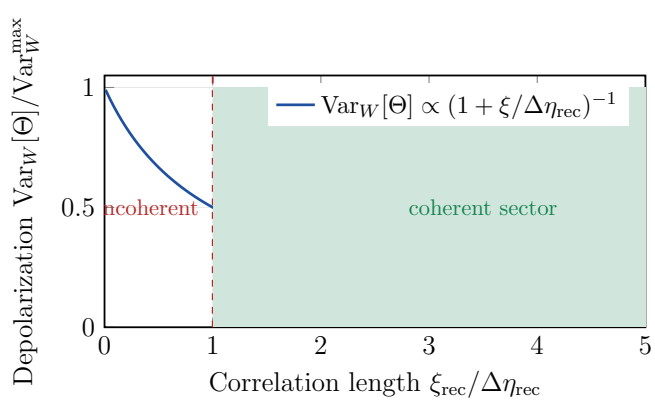

Figure 4: Depolarization $\mathrm{Var}_W[\Theta]$ as a function of the ratio $\xi_{\rm rec}/\Delta\eta_{\rm rec}$. The washout bound requires $\xi_{\rm rec} \gtrsim \Delta\eta_{\rm rec}$ (green region). The boundary at $\xi_{\rm rec} = \Delta\eta_{\rm rec}$ marks the transition from coherent rotation to dominant depolarization.

The washout observational bound therefore constrains not a particle mass but the axial synchronization length of the plenum at $z \approx 1100$ [31, 32].

7 Continuous Axial Phase Synchronization and the Coherence Theorem

7.1 XY synchronization model

The discrete Ising model captures combinatorial synchronization but suppresses the essential S^1 geometry of polarization. We replace it with an anisotropic XY model [31] with a continuous phase field $\theta(x) \in S^1$. In the continuum limit the free energy functional is

$$\mathcal{F}[\theta] = \int d^4x\, dn_4 \left[\frac{K}{2}(\partial_\mu \theta)^2 + \frac{K_4}{2}(\partial_{n_4}\theta)^2 - \lambda\, \Xi(x) \cos\theta \right], \tag{30}$$

with Ξ the plenum pseudoscalar (8). The axial projection is $\kappa_A = \kappa_0 \partial_\eta \theta$, recovering the boundary structure $\beta(\eta) = \kappa_0[\theta(\eta_0) - \theta(\eta)]$.

7.2 Correlation function and variance

Decomposing $\theta = \bar{\theta} + \delta\theta$ with $\langle \delta\theta \rangle = 0$, the suppression factor is

$$|\mathcal{W}_\pm| = \exp\!\big(-2\kappa_0^2 \mathrm{Var}_W[\delta\theta]\big)\,, \tag{31}$$

with

$$\mathrm{Var}_W[\delta\theta] = \int d\eta\, d\eta'\, W(\eta) W(\eta') \langle \delta\theta(\eta)\delta\theta(\eta') \rangle. \tag{32}$$

For an exponential two-point function $\langle \delta\theta(\eta)\delta\theta(\eta') \rangle \propto e^{-|\eta-\eta'|/\xi_{\mathrm{rec}}}$:

$$\mathrm{Var}_W[\delta\theta] \sim \frac{\Delta\eta_{\mathrm{rec}}}{\xi_{\mathrm{rec}}} \quad (\xi_{\mathrm{rec}} \ll \Delta\eta_{\mathrm{rec}}), \tag{33}$$

$$\mathrm{Var}_W[\delta\theta] \sim \mathrm{const} \quad (\xi_{\mathrm{rec}} \gg \Delta\eta_{\mathrm{rec}}). \tag{34}$$

7.3 The Coherence Theorem

Theorem 7.1 (Coherence Bound on the Parity-Odd Plenum Sector)**.** *Let $\kappa_A(\eta) = \kappa_0 \partial_\eta \theta(\eta)$ be the axial projection of the RSVP connection along null geodesics, where θ has correlation length ξ_{rec} across the recombination visibility kernel of width $\Delta\eta_{\mathrm{rec}}$.*

(i) Coherent regime. *If $\xi_{\mathrm{rec}} \gg \Delta\eta_{\mathrm{rec}}$, then $\beta_{\mathrm{obs}} \approx \kappa_0(\theta(\eta_0) - \theta(\eta_\star))$ with negligible washout.*

(ii) Incoherent regime. *If* $\xi_{\rm rec} \ll \Delta\eta_{\rm rec}$*, then polarization amplitude is suppressed by* $|\mathcal{W}_\pm| = \exp(-2\kappa_0^2 \mathrm{Var}_W[\delta\theta])$ *with* $\mathrm{Var}_W[\delta\theta] \propto \Delta\eta_{\rm rec}/\xi_{\rm rec}$.

(iii) Observational bound. *The empirical constraint* $|\mathcal{W}_\pm| \geq 1 - \epsilon_{\rm obs}$ *[5, 10] implies*

$$\boxed{\xi_{\rm rec} \gtrsim \frac{2\kappa_0^2 \Delta\eta_{\rm rec}}{\epsilon_{\rm obs}}.} \tag{35}$$

Proof. Parts (i) and (ii) follow from inserting the exponential correlation model into (32) and evaluating the asymptotic limits. Part (iii) follows by combining (31) with $|\mathcal{W}_\pm| \geq 1 - \epsilon_{\rm obs}$, which gives $\mathrm{Var}_W[\delta\theta] \leq \epsilon_{\rm obs}/(2\kappa_0^2)$, and substituting the scaling (33). □

Remark 7.2. Inequality (35) is a *geometric observational bound on the coherence of the parity-odd plenum sector at recombination.* It does not specify the microphysical origin of θ but constrains its correlation structure, replacing particle-exclusion statements with a synchronization-length bound.

8 Entropy–Coherence Inequality and Implications for Computing

8.1 Phase-coherent transport as holonomy

The coherence theorem identifies a universal structure: controlled holonomy across a finite integration kernel requires $\mathrm{Var}_W[\delta\Theta] \lesssim \varepsilon$. This structure appears in any system in which computation is performed through controlled phase transport, with the cosmological and engineering identifications

$$\eta \leftrightarrow t, \quad W(\eta) \leftrightarrow K(t), \quad \kappa_A(\eta) \leftrightarrow \Omega(t), \tag{36}$$

where $K(t)$ is the detector/device integration kernel and $\Omega(t)$ the phase rotation rate [42–44].

8.2 The entropy–coherence inequality

For a synchronization field with diffusion coefficient $D_\theta = k_B T/\gamma$ (Einstein relation [37]), the phase variance over a kernel of width Δt scales as $\mathrm{Var}_W[\delta\Theta] \sim 2D_\theta \Delta t$. The coherence condition becomes

$$\frac{k_B T}{\gamma} \Delta t \lesssim \varepsilon, \tag{37}$$

or equivalently

$$\boxed{\gamma \geq \frac{k_B T \Delta t}{\varepsilon}.} \tag{38}$$

The entropy export rate (through damping γ) must exceed a threshold proportional to temperature and integration time in order to maintain coherent phase rotation [38, 39].

8.3 Multi-modal architectures

For N modes with independent diffusion coefficients D_a, $\mathrm{Var}_W = \sum_a D_a \Delta t$ while the coherent mean scales as $\sum_a \bar{\Theta}_a$. The multi-modal entropy–coherence inequality is

$$\sum_{a=1}^{N} \frac{k_B T_a}{\gamma_a} \Delta t \lesssim \varepsilon, \tag{39}$$

which may be satisfied with weaker individual damping when modes are independent. This is the thermodynamic statement of the $\sqrt{N}$ signal-to-noise improvement (22): distributing the phase holonomy across N weakly coupled channels requires proportionally less entropy export per channel [40].

	Cosmological	Photonic/thermodynamic
Phase variable	$\theta(\eta)$ (axial plenum)	$\theta(t)$ (optical mode)
Rotation rate	κ_A	$\Omega(t)$
Integration kernel	$W(\eta)$ (visibility)	$K(t)$ (detector window)
Kernel width	$\Delta\eta_{\mathrm{rec}}$	Δt
Coherence bound	$\xi_{\mathrm{rec}} \gg \Delta\eta_{\mathrm{rec}}$	$\xi_c \gg \Delta t$
Entropy constraint	$\gamma \geq k_B T \Delta\eta_{\mathrm{rec}}/\varepsilon$	$\gamma \geq k_B T \Delta t/\varepsilon$

Table 1: Structural correspondence between the cosmological washout constraint and the engineering coherence limit for phase-based computation. Both are governed by the same entropy–coherence inequality (38).

9 Predictive Discriminants: Geometric Coherence versus Free Pseudoscalar

The geometric interpretation and the minimally coupled pseudoscalar model agree at the level of isotropic mean rotation and uniform washout. They diverge in four independently testable observational predictions.

9.1 Anisotropic birefringence correlations with large-scale structure

In the axion model with homogeneous initial conditions, $\langle \beta(\hat{n})\mathcal{O}(\hat{n}')\rangle = 0$ for any large-scale structure observable $\mathcal{O}$ unless the ALP couples directly to density perturbations [12, 51].

In RSVP, $\kappa_A \propto \nabla S \cdot \omega$, so $\beta(\hat{n})$ is sourced by entropy gradients and vorticity along the line of sight. This predicts

$$C_\ell^{\beta\,\mathrm{kSZ}} \neq 0, \qquad C_\ell^{\beta\,\nabla S} \neq 0, \tag{40}$$

where kSZ denotes the kinetic Sunyaev–Zel'dovich signal [52, 53] and ∇S is traced by, e.g., large-scale peculiar velocity maps or void ellipticity statistics [54, 55]. The anisotropic birefringence power spectrum [49, 50] acquires contributions

$$C_\ell^{\beta\beta} \sim \int d\chi\, W_\chi^2 \langle (\nabla S \cdot \omega)(\chi\hat{n})(\nabla S \cdot \omega)(\chi\hat{n}')\rangle, \tag{41}$$

which reflect the geometric structure of the plenum.

9.2 Multipole-dependent depolarization

In the simplest homogeneous axion model, the washout factor $\mathcal{W} = J_0(g_{\phi\gamma}\phi_\star)$ is uniform across multipoles ℓ to leading order.

In RSVP, if the axial sector has scale-dependent coherence $\xi(\ell) \propto \ell^{-\gamma}$, the suppression factor acquires ℓ-dependence:

$$|\mathcal{W}_\ell| = \exp\Big(-2\kappa_0^2 \mathrm{Var}_W^{(\ell)}[\delta\theta]\Big)\,, \tag{42}$$

producing measurable scale-dependent amplitude suppression in CMB E- and B-modes [10, 49, 56]. Future experiments such as CMB-S4 [57] and LiteBIRD [58] will have sufficient sensitivity to detect or constrain ℓ-dependent depolarization.

9.3 Cross-correlation with redshift-space entropy gradient

In the RSVP Everlasting Redshift framework, $z(\hat{n}) \propto \int_\gamma \nabla_\mu S\, dx^\mu$. Combined with the birefringence source $\kappa_A \propto \nabla S \cdot \omega$, the cross-spectrum

$$\langle \beta(\hat{n})\, z(\hat{n}')\rangle \neq 0 \tag{43}$$

at second order in perturbations. No such correlation is expected in the canonical ALP model. A statistically significant detection of (43) would constitute a unique signature of geometric axial coherence.

9.4 Non-Gaussian phase statistics

The axion model predicts Gaussian $\beta(\hat{n})$ statistics for Gaussian initial conditions. A synchronization-based sector near criticality generically produces non-Gaussian phase fluctuations with

$$\langle \beta^3 \rangle_c \neq 0, \qquad \langle \beta^4 \rangle_c \propto \xi^{4-d}, \tag{44}$$

where d is the effective dimensionality [32, 33]. Measurement of a nonzero birefringence bispectrum [50] provides a further discriminant.

Observable	ALP model	RSVP geometric
Isotropic $\bar{\beta}$	✓	✓
$\lvert\mathcal{W}_\pm\rvert$ **suppression**	✓	✓
ℓ**-independent washout**	✓	×
$C_\ell^{\beta\,\mathrm{kSZ}} = 0$	✓	×
$C_\ell^{\beta\,z} = 0$	✓	×
Gaussian β **statistics**	✓	(near-critical: ×)

Table 2: Comparison of observational predictions. Entries marked ✓ are expected; × marks a prediction that differs. The two interpretations are degenerate only for the first two rows.

10 Concluding Synthesis: Axial Holonomy as a Coherence Principle

We have developed a reinterpretation of cosmic birefringence within the Relativistic Scalar–Vector Plenum framework in which the observable CMB polarization rotation is the axial holonomy of a parity-odd geometric sector, not the displacement of a propagating pseudoscalar. The central results are:

1. The RSVP plenum connection contains a parity-odd sector $\Delta^{(\mathrm{odd})}$ sourced by entropy–vorticity coupling (8), whose null-ray projection κ_A generates polarization holonomy (5).

2. The observable rotation is *not* $H_A[\gamma]$ but its convolution with the recombination visibility kernel, yielding suppression factor $|\mathcal{W}_\pm|$ governed by phase variance of κ_A across $\Delta\eta_{\mathrm{rec}}$.

3. The washout bound is equivalent to the coherence inequality (16) and, via theorem 7.1, to a lower bound on the axial correlation length ξ_{rec} at $z \approx 1100$.

4. Multi-modal factorization (14) universally improves the signal-to-washout ratio as $\sqrt{N}$ through variance additivity; the Shao–Obata–Zhang two-field mechanism [1] is the $N = 2$ instance.

5. The entropy–coherence inequality (38) unifies the cosmological washout bound with the thermodynamic cost of phase-stable optical computation.

6. Four classes of observational discriminants (table 2) distinguish geometric axial coherence from free pseudoscalar transport, accessible to CMB-S4 [57] and LiteBIRD [58].

The conceptual shift is precise. The question is no longer "which particle produces β?" but "what geometric property of the plenum renders the parity-odd sector sufficiently coherent across recombination to generate the observed holonomy?" In RSVP, geometric ordering competes with entropy production: coherent axial holonomy represents ordered structure in the parity-odd sector; washout represents entropy-dominated phase diffusion. The CMB therefore encodes a constraint on the entropy content of the universe's parity-odd geometric degrees of freedom at recombination—a cosmological instance of the universal entropy–coherence principle (table 1).

The remaining task is empirical. Should future observations reveal anisotropic correlations, scale-dependent depolarization, or non-Gaussian phase statistics in the birefringence field, the geometric coherence interpretation will gain support. In either case, the coherence theorem clarifies what is being tested: not merely the existence of a field, but the structure of phase synchronization in the parity-odd sector of the cosmos.

Appendix A Optical Transport with Axial Connection Component

Let k^μ be a null vector with $k^\mu k_\mu = 0$, $k^\nu \nabla_\nu k^\mu = 0$, and e^μ_a $(a = 1, 2)$ orthonormal polarization vectors satisfying $e^\mu_a k_\mu = 0$, $e^\mu_a e_{b\mu} = \delta_{ab}$ [26]. Define an effective connection

$$\widetilde{\nabla}_\nu e^\mu_a = \nabla_\nu e^\mu_a + \mathcal{A}_\nu \epsilon_{ab} e^\mu_b, \qquad \kappa_A = \mathcal{A}_\mu k^\mu. \tag{45}$$

Parallel transport $k^\nu \widetilde{\nabla}_\nu e^\mu_a = 0$ gives $d\psi/d\eta = \kappa_A(\eta)$, hence $\beta = \int_\gamma \kappa_A \, d\eta$.

Appendix B Visibility-Weighted Holonomy: Detailed Derivation

With $\beta(\eta) = \int_\eta^{\eta_0} \kappa_A d\eta'$ and $\beta_{\rm obs} = \int W(\eta)\beta(\eta)d\eta$, interchange integration order:

$$\beta_{\rm obs} = \int \kappa_A(\eta') \left(\int_{\eta_{\rm LSS}}^{\eta'} W(\eta) d\eta \right) d\eta' = \int \kappa_A(\eta') \mathcal{F}(\eta') d\eta'. \tag{46}$$

For $\mathcal{F}(\eta') \approx \Theta(\eta' - \eta_\star)$ (step function approximation), $\beta_{\rm obs} \approx \int_{\eta_\star}^{\eta_0} \kappa_A d\eta'$.

Appendix C Phase Variance and Depolarization

Define $\Theta = 2\beta$. The characteristic function $\mathcal{W}_\pm = \langle e^{\pm i\Theta} \rangle_W$ satisfies, under Gaussian fluctuations, $|\mathcal{W}_\pm| = \exp\left(-\frac{1}{2}\mathrm{Var}_W[\Theta]\right)$. For stationary correlations $\langle \kappa_A(\eta)\kappa_A(\eta') \rangle \propto e^{-|\eta-\eta'|/\xi}$:

$$\mathrm{Var}_W[\Theta] \sim \frac{\Delta\eta_{\rm rec}}{\xi} \quad (\xi \ll \Delta\eta_{\rm rec}), \qquad \mathrm{Var}_W[\Theta] \sim \text{const} \quad (\xi \gg \Delta\eta_{\rm rec}). \tag{47}$$

Appendix D Multi-Modal Variance Additivity

Let $\kappa_A = \sum_{a=1}^N \kappa_A^{(a)}$ with $\langle \kappa_A^{(a)} \kappa_A^{(b)} \rangle = 0$ for $a \neq b$. Then $\mathrm{Var}_W[\Theta] = \sum_a \mathrm{Var}_W[\Theta_a]$ and $\langle \beta \rangle = \sum_a \langle \beta_a \rangle$, giving $\langle \beta \rangle / \sqrt{\mathrm{Var}_W} \propto \sqrt{N}$ [45, 46].

Appendix E Renormalization Group Analysis

The Euclidean action $S[\theta] = \int d^d x [\frac{K}{2}(\nabla\theta)^2 - \lambda \Xi \cos\theta]$ with dimensionless couplings $g = T/K$ and $y = \lambda/T$ admits one-loop beta functions for $d = 4 - \epsilon$:

$$\beta_g = -\epsilon g + C_d g^2 + \mathcal{O}(g^3), \qquad \beta_y = \left(2 - \frac{g}{4\pi^2}\right) y + \mathcal{O}(g y^2). \tag{48}$$

The Wilson–Fisher fixed point $g^* = \epsilon/C_d$ yields $\xi \sim |g - g^*|^{-\nu}$ with $\nu = \frac{1}{2} + \mathcal{O}(\epsilon)$ [32–34]. The washout bound $\xi_{\rm rec} \gg \Delta\eta_{\rm rec}$ requires $|t| \lesssim (\Delta\eta_{\rm rec}/L_0)^{-1/\nu}$ where $t = (g - g^*)/g^*$.

Appendix F Stochastic Phase Dynamics (Langevin/MSR)

The Langevin equation $\partial_\eta \theta = -\Gamma \delta\mathcal{F}/\delta\theta + \zeta$ with $\langle \zeta\zeta' \rangle = 2\Gamma T \delta\delta$ has the mode solution

$$\langle \theta_k(\eta)\theta_{k'}(\eta') \rangle = (2\pi)^3 \delta(\mathbf{k} + \mathbf{k}') \frac{T}{Kk^2 + m^2} e^{-\Gamma(Kk^2+m^2)|\eta-\eta'|}, \tag{49}$$

giving correlation length $\xi = m^{-1}$ and coherence time $\tau_c = (\Gamma m^2)^{-1}$ [38, 41]. The visibility-weighted variance is

$$\mathrm{Var}_W[\Theta] = 4\kappa_0^2 \int \frac{d^3k}{(2\pi)^3} \frac{T}{Kk^2 + m^2} |\widetilde{W}(k)|^2, \tag{50}$$

where $\widetilde{W}$ is the Fourier transform of $W(\eta)$. The coherence condition $\tau_c \gg \Delta\eta_{\rm rec}$ is equivalent to $\Gamma m^2 \ll \Delta\eta_{\rm rec}^{-1}$.

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